

4.11 Creating Queries in MS Access

کوئری ٹیبل کو بدلے بغیر سوال پوچھتی ہے۔

CS XI

Chapter 4: Data and Analysis

Lecture 13 of 15

★ Golden

4.11 Creating Queries in MS Access

★ Golden

LEARN IT

A query retrieves, filters and analyses data **without changing the original tables**. A simple SELECT query:

1. Create tab → Query Design.
2. Show Table → add Student → Close.
3. Double-click the fields you want; they drop into the grid.
4. Red Run (!) button executes it.

Criteria restrict rows. Type a value in the Criteria row under a field.

Goal	Where to type	Example
Single criterion	Criteria row, one column	Department → "Computer Science"
AND (both must match)	Same Criteria row, two columns	Name "Ali" and Department "Computer Science"
OR (either may match)	Criteria row and the or: row	Department "IT" or "Computer Science"

Wildcards in Access: * = any characters, ? = one character. Like "A*" finds Ali, Ahmed, Ayesha.

WORKED EXAMPLE

Equivalent SQL (what Access writes behind the grid):

```
SELECT StudentName, Department
FROM Student
WHERE Department = "Computer Science";

SELECT StudentName, Department
FROM Student
WHERE StudentName = "Ali" AND Department = "Computer Science";

SELECT StudentName, Department
FROM Student
WHERE Department = "IT" OR Department = "Computer Science";
```

EXAM TIP

Remember: AND = same row in the grid; OR = different rows. Examiners draw a blank grid and ask you to fill Criteria and or:.

COMMON MISTAKE

Text criteria need quotes: "IT". Numbers do not: ≥ 80 . Mixing these is a common cause of empty results.

اردو میں سمجھیں

کوٹری ٹیبل کو بدلے بغیر سوال پوچھتی ہے۔ ایک شرط Criteria قطار میں، AND ایک ہی قطار کی دو کالموں میں، OR الگ قطار (or:) میں۔ وائلڈ کارڈ * کئی حروف، ؟ ایک حرف۔

CHECK YOURSELF

1. Criteria for marks at least 80?
2. You want CS or IT. Same row or two rows?

Answers: 1. ≥ 80 under Marks. 2. Two rows (OR).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A query is:

- a. A request to retrieve/filter/calculate data
- b. A printed letter
- c. A cable
- d. A gate

MCQ 2. Criteria in a query:

- a. Filter rows (e.g. Marks $\geq$ 40)
- b. Change field types
- c. Draw ER
- d. Set the OS

MCQ 3. A parameter query asks:

- a. The user for a value at run time
- b. The CPU voltage
- c. TCP window
- d. Figma frames

MCQ 4. Select queries by default:

- a. Do not delete stored rows
- b. Always wipe the table
- c. Always cascade
- d. Always print

MCQ 5. Sorting in a query:

- a. Orders the result (Asc/Desc)
- b. Changes PK
- c. Encrypts
- d. Creates a form

MCQ 6. This topic is ★ because board papers love:

- a. Writing criteria and showing results
- b. Cabling
- c. NAND only
- d. Sprints

Extra short questions

1. Define query.
2. Write criteria: CS students with marks < 40.
3. Difference between filter on a datasheet and a query.
4. What is a calculated field in a query?

Extra long question

1. Design a query: join Students and Marks, show Fail list (marks<40) sorted by name, with a calculated column 40-Marks as Needed. Write QBE grid/SQL-like criteria. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A saved question that selects, filters, sorts, calculates from tables.
2. Class = "CS" AND Marks < 40 (field names as in the table).
3. Query is reusable, can join tables, calculate, feed reports.
4. Total:[Math]+[Eng] — not stored, computed.

Long

1. Tables joined on roll. Fields Name, Subject, Marks, Needed: 40-[Marks]. Criteria Marks<40. Sort Name. Result sample two rows.

Next lecture: 4.11.4 Data Summarization & Statistical Analysis